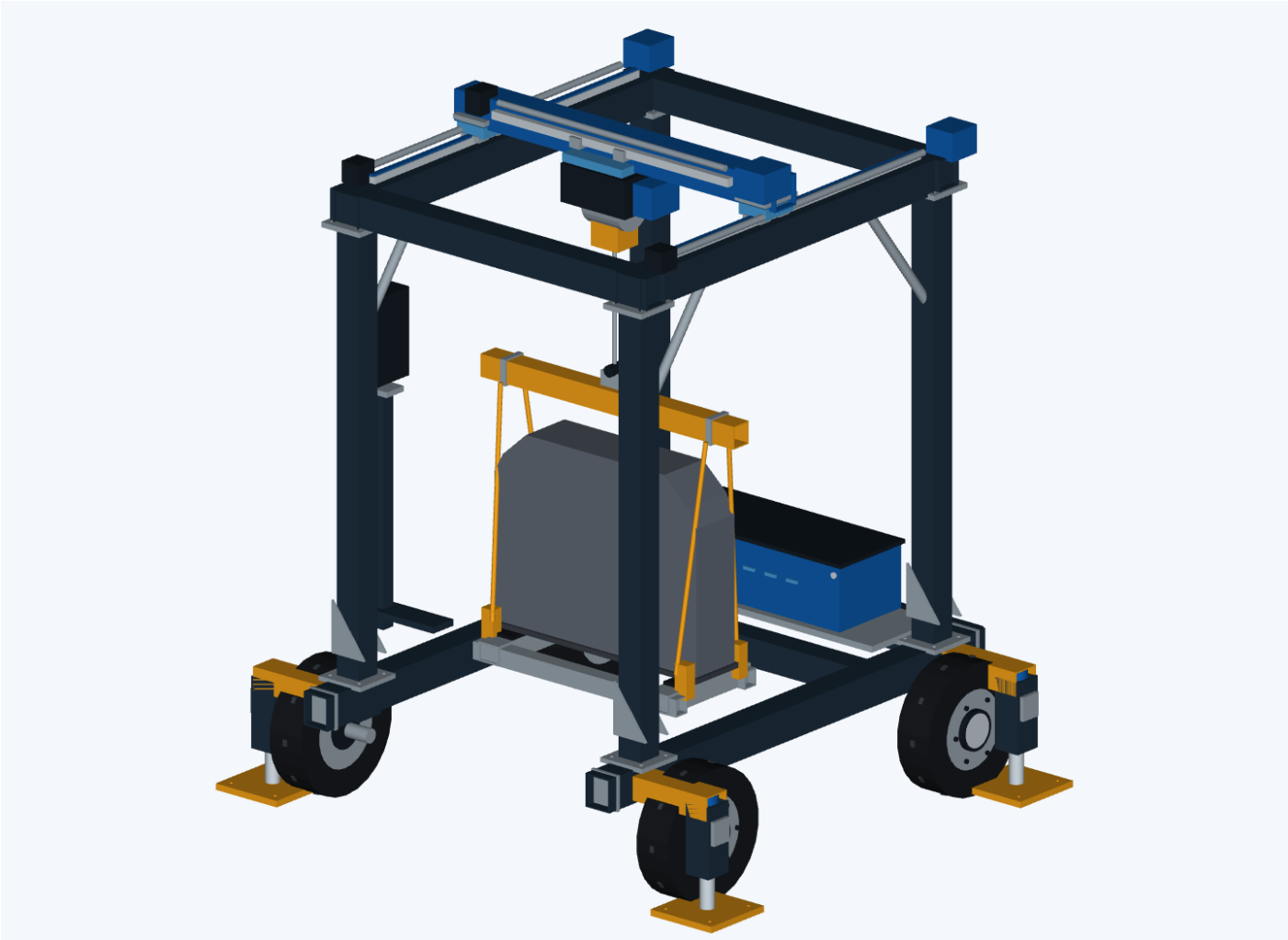
Machine design



Current CAD export showing the mobile frame, top carriage, wheel assemblies and stabilizer arrangement.

Principal design dimensions

Component	CAD geometry
Chassis	1,360 mm wide × 1,950 mm long
Wheel-center spacing	1,536 mm across × 1,290 mm fore-aft
Wheel diameter	450 mm
Gantry top	2,050 mm

PROTOTYPE 3 / CAD VIEW

Frame and test-stone geometry



Existing design frame shown with the smaller stone geometry. Components omitted from this view are documented in the full CAD.

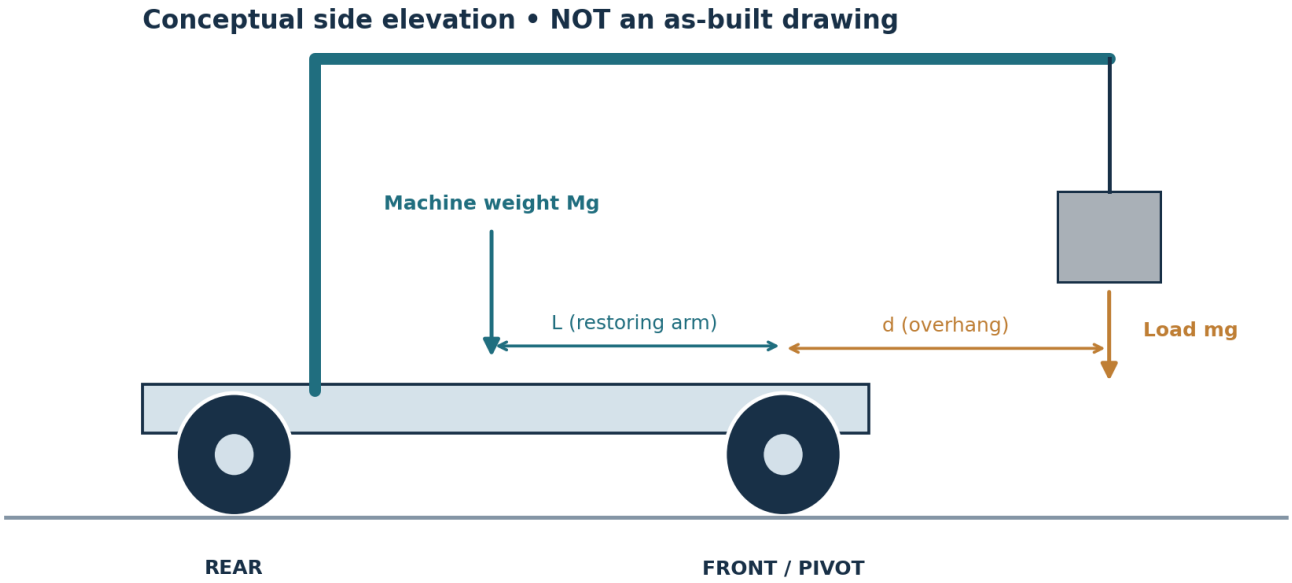
Test-stone dimensions

Dimension	Approximate original stone
Left-right, across chassis	16 in / 406.4 mm
Front-to-back	8 in / 203.2 mm
Height	6 in / 152.4 mm
Mass	Approximately 70 lb

This CAD view provides the geometry and orientation. The exact attachment point and load overhang during the physical test were not recorded.

DESIGN AND MECHANICS

Relationship to the stability model



Idealized front-pivot force model used for the longitudinal calculation.

Load-position effect

Moving the vertical suspension line farther ahead of the front wheel-contact line increases the load moment mgd. In the simple static equilibrium, the rear reaction is:

$$N_{rear} = (M g L - m g d) / B$$

Different kinds of information

CAD	Physical test	Calculation
Design geometry and arrangements	Observed rear-wheel lift and pitching	Static force predictions for declared assumptions

Interpretation

The digital drawings show machine geometry and design intent. The physical test describes observed behavior, but does not supply the original reaction forces. The mathematical model examines load-position sensitivity; it is not an as-built load certification.

The simplified stone position is a CAD presentation choice, not a recovered physical test coordinate.